

SIH26155 - Network Security Compliance Auditor

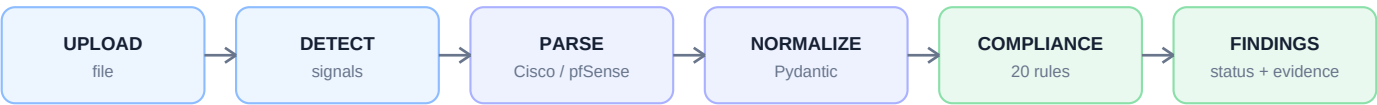
Architecture Document | TrailBlazers | Smart India Hackathon 2026 | Current merged MVP

PS	THEME	VENDORS	CORE	TEAM
SIH26155	Blockchain & Cybersecurity	Cisco IOS + pfSense	20 CIS-aligned themes	TrailBlazers

1. System objective

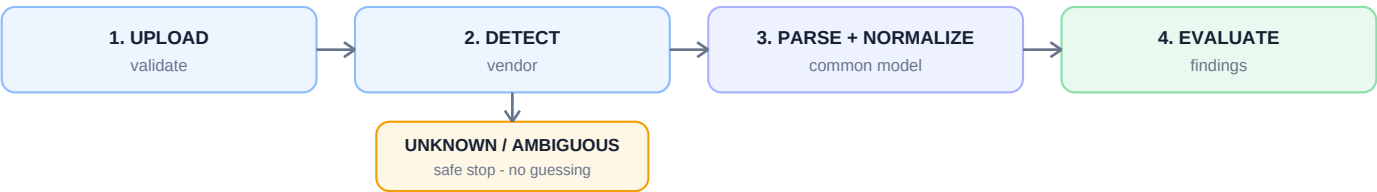
The current merged MVP accepts a single configuration file, detects Cisco IOS or pfSense, parses vendor syntax into a common Pydantic model, evaluates applicable CIS-aligned security themes deterministically, preserves evidence, reports severity, and provides remediation guidance. Insufficient evidence can produce NOT_ASSESSED instead of an invented result.

2. High-level system architecture



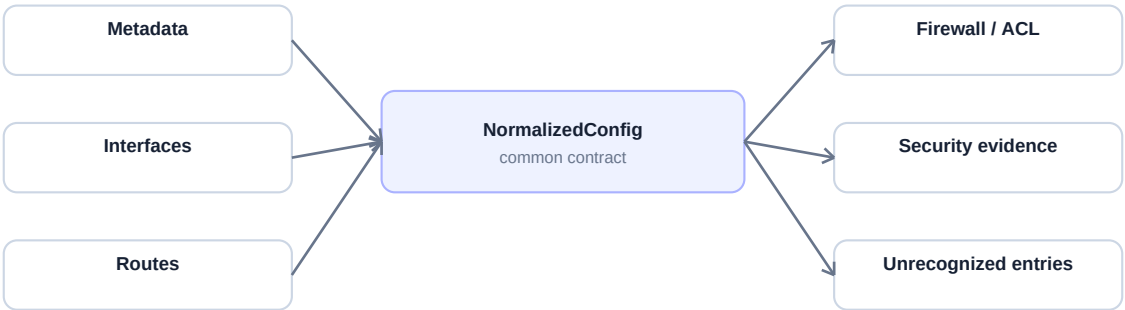
Boundary: parsers extract facts; compliance decides status; AI only assists semantic mapping.

3. Audit pipeline and safe-stop behavior



STAGE	OUTPUT
Upload	Raw configuration
Detect	Vendor + confidence + signals
Parse / Normalize	NormalizedConfig
Evaluate	Findings + status
Report	Audit output / PDF

4. Vendor-neutral data model

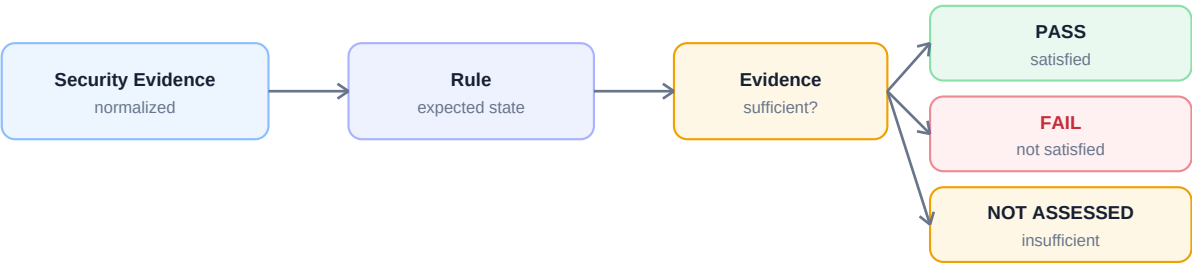


Common model: compliance rules consume normalized security properties instead of raw vendor syntax.

Architecture - Decision, Learning, and Analysis Layers

Deterministic core | human-in-the-loop learning | ordered-rule analysis

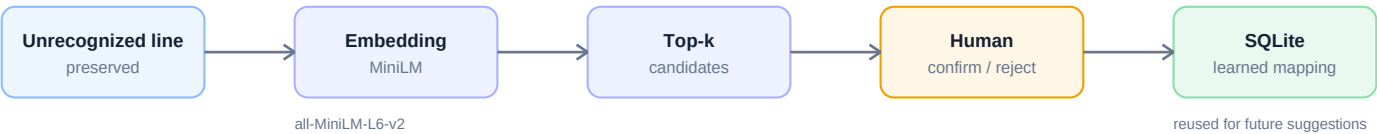
5. Deterministic compliance engine



Internal IDs CIS-NET-01 ... CIS-NET-20 are application identifiers, not official CIS recommendation numbers.

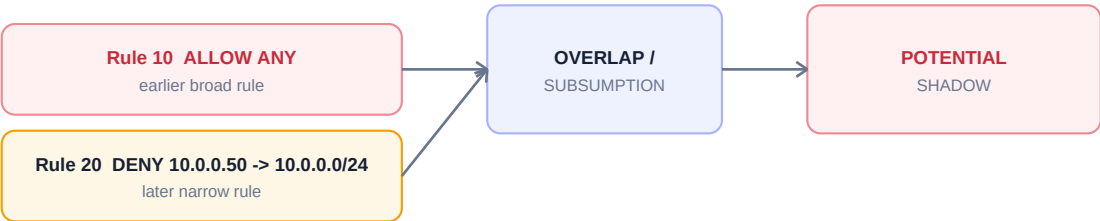
STATE	OPERATIONAL MEANING
PASS	Evidence satisfies the expected state.
FAIL	Evidence contradicts the control, or the rule defines missing required configuration as a failure.
NOT_ASSESSED	Reliable evidence is insufficient; the engine does not invent an answer.

6. AI-assisted semantic learning



PROPERTY	BEHAVIOR
Model	Sentence Transformers: all-MiniLM-L6-v2
Candidates	Ranked top-k mappings with similarity and confidence band
Persistence	Human confirmation required before SQLite storage
Authority	AI suggests; deterministic compliance decides

7. Shadow rule detection



A broader earlier rule can make a narrower later rule ineffective under first-match processing. The merged implementation uses an MVP overlap / subsumption analyzer; it is not a complete packet-processing equivalence engine.

Design rule: deterministic evidence remains the compliance authority; AI and shadow analysis are advisory layers.

Architecture - Evidence, Security, Technology, and Scope

Audit evidence | failure behavior | delivery stack | current and deferred capabilities

8. Evidence and reporting

AREA	ARCHITECTURE RESPONSIBILITY
Evidence	Trace findings to observed configuration evidence and expected state.
Severity	Prioritize review without hiding the underlying evidence.
Remediation	Provide ready-to-copy recommendations; no automatic execution.
Reporting	ReportLab PDF generator consolidates metadata, findings and remediation.
Rule analysis	Shadow / ordering observations are surfaced where available.

9. Security and failure behavior

BOUNDARY	BEHAVIOR
Unknown vendor	Never force-classified.
Missing evidence	NOT_ASSESSED where appropriate.
AI	Advisory only.
Learning	Human confirmation required.
Remediation	Recommendation only; no automatic execution.
Similarity	Ranking signal, not probability.
Vendor logic	Isolated in parser / normalization layers.

10. Technology stack

LAYER	TECHNOLOGY
UI	React + Vite
API	FastAPI + Uvicorn
DATA	Pydantic SQLite
PARSING	Regex (Cisco IOS) XML (pfSense)
COMPLIANCE	JSON rules deterministic Python
LEARNING	Sentence Transformers all-MiniLM-L6-v2
REPORT / QA	ReportLab Pytest

11. Current scope and deferred work

CURRENT MVP

- Single-file audit
- Cisco IOS + pfSense
- Vendor detection + normalization
- 20 CIS-aligned themes
- PASS / FAIL / NOT_ASSESSED
- Evidence + severity + remediation
- Semantic learning + shadow analysis
- PDF report generator

DEFERRED / FUTURE

- FortiGate
- NIST implementation
- Bulk inventory / batch uploads
- Live SSH
- Automatic remediation
- RBAC / chatbot
- Change-impact analysis
- Blockchain / tamper-evident ledger

Evaluation principle: deterministic evidence first; AI assists mapping, not compliance decisions.